

Positive Square Energy of Every Decacyclic Cactus

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Abstract

Let G be a connected cactus of cyclomatic rank ten on n vertices. We prove $s^+(G) > n$. The sharp block-additive doubly nonnegative estimate reduces the problem exactly to T^9Q and T^8PP , where $T = C_3$, $P = C_5$, and Q is an arbitrary cycle. Exact colored cluster-partition censuses give $96 = 92 + 4$ and $180 = 170 + 10$ proper rows. Exhaustive reduced-tree pruning and marked finite certificates reduce their structural rows to $A_9 \mid Q$, $T^8P \mid P$, and $P \mid A_8 \mid P$. The first has $3624 = 3621 + 3$ marked-entry classes; the second has $11689 = 11586 + 100 + 3$; and the third has $11689 = 11674 + 15$. In the one-cluster case the full T^9Q census has at most 8049 color-preserving incidences and an exact three-row hostile frontier. The T^8PP census is $30386 = 30377 + 9$, and all nine exceptions have explicit positive replacements. Separations use actual bridges or consecutive cycle intervals, and every connector remnant, incidence branch, shared cut, and attached tree receives exactly one final owner. A tempting packing-one guard is false on the new three-cut kernels; we instead open a leaf cycle and apply packing one only to the retained common-hub packet. No open two-pivot winding assertion is used.

1 Definitions, inputs, and firewall

All graphs are finite, simple, and undirected. A cactus is a connected graph whose blocks are bridges or cycles; its cyclomatic rank is the number of cyclic blocks. For adjacency eigenvalues $\lambda_1, \dots, \lambda_n$, set

$$s^+(G) = \sum_{\lambda_i > 0} \lambda_i^2, \quad s^-(G) = \sum_{\lambda_i < 0} \lambda_i^2, \quad \sigma(G) = s^+(G) - |V(G)|.$$

For every induced partition $V(G) = V_1 \dot{\cup} \dots \dot{\cup} V_r$, positive square energy is superadditive:

$$s^+(G) \geq \sum_i s^+(G[V_i]), \quad \sigma(G) \geq \sum_i \sigma(G[V_i]). \quad (1)$$

This is the theorem of Akbari, Kumar, Mohar, and Pragada [1]. Since a connected rank-ten graph has $|E| = n + 9$, the theorem below proves the strict positive-square-energy conjecture for this class.

Write $T = C_3$, $P = C_5$, and

$$\delta = \sec(\pi/5) - 1 = \sqrt{5} - 2 < \frac{1}{4}, \quad \delta_q = \sec(\pi/q) - 1 < 1 \quad (q \geq 4).$$

A cycle $Q = C_q$ is *hostile* when $q \equiv 1 \pmod{4}$. We use the proved rank-nine cactus theorem, including arbitrary attached trees [4], the common-cut and one-hostile-cycle packet theorems [14, 13], and the exact rank-ten artifacts listed in Appendix A.

The proof has a strict firewall. It does not use the proposed two-pivot phase or winding assertion of [16]; the exact two-terminal reduction there does not prove the needed integrated argument

bound. It does not infer packing number one from a nearest-cycle or Voronoi territory. It does not contract a bridge connector analytically, split a retained shared cut between owners, use edge monotonicity, or replace a cyclic argument by a principal arctan. Every invocation of packing one below displays the actual common hub of the retained triangles. The two new apparent two-pivot rows are closed by opening a leaf cycle at exact tree cost -1 .

2 Packets, cycle order, and ownership

The rank-nine theorem and its packet infrastructure give, uniformly over arbitrary finite trees attached at arbitrary packet vertices,

$$\sigma(H) \geq 0 \quad \text{for connected ranks two and three,} \quad (2)$$

$$\sigma(H) > 0 \quad \text{for connected ranks four through nine,} \quad (3)$$

$$\sigma(P) \geq -\delta, \quad \sigma(Q) \geq -\delta_q, \quad (4)$$

$$\sigma(TP) > 1 - \delta, \quad \sigma(TQ) > 0, \quad \sigma(TTQ) \geq 0, \quad (5)$$

$$\sigma(PP) > 0, \quad \sigma(TPP) > \frac{3}{2}. \quad (6)$$

For a connected shared-cut cluster A_r of r triangles we use

$$\sigma(A_r) > b_r, \quad (b_1, \dots, b_8) = (0, 1, 2, 3, 2, 1, 0, 0). \quad (7)$$

Lemma 2.1 (One-hostile-cycle packing one). *Let the cyclic blocks of a connected cactus H be one hostile $Q = C_q$ and $a \geq 1$ triangles, no two of which are vertex-disjoint. The Q may meet the triangular part at its root or be joined to it by an arbitrary positive path, and arbitrary trees may occur everywhere. Then*

$$s^+(H) - s^-(H) > -2\delta_q, \quad \sigma(H) > a - \delta_q.$$

Proof. Eliminate every tree off the cyclic spine by its nonnegative matching message. In the grouped Sachs expansion on it, write $\Psi_H(t) = R + 2i(B - A)$, where B is the matching partition after deleting Q , and A is the sum after additionally deleting one triangle. Packing one excludes every term containing two triangles. Splitting matchings by use of the Q boundary gives

$$R - Z_q(t)(B - A) = E + LB + Z_q(t)A + 4 \sum_{T \cap Q = \emptyset} Z_{H-(V(Q) \cup V(T))} > 0.$$

Thus the continuous argument is strictly smaller than the isolated hostile cycle argument. The signed Coulson identity gives the first inequality. Since $|E(H)| = |V(H)| + a$ and $s^+ + s^- = 2|E|$, the second follows. This is the standalone argument of [13], not a two-pivot estimate. $\square$

Lemma 2.2 (Common-cut packets). *For a one-vertex bouquet, with arbitrary attached trees,*

$$\sigma(T^k Q) > k - \delta_q \quad (q \equiv 1 \pmod{4}), \quad (8)$$

$$\sigma(T^k Q) > k \quad (q \text{ even or } q \equiv 3 \pmod{4}), \quad (9)$$

$$\sigma(T^k P) > k - \delta, \quad (10)$$

$$\sigma(T^k PP) > k + 1 - \frac{4}{3\sqrt{13}}. \quad (11)$$

Proof. After exact tree elimination, the normalized determinant is the scalar sum

$$W(t) = a_x + \sum_j \frac{B_j + \omega_{\ell_j}}{A_j}, \quad \omega_{\ell} = -2i^{-\ell},$$

with $A_j, B_j > 0$. Comparing its real and imaginary parts with the isolated hostile lobe gives (8)–(10). For two pentagons the exact coefficient inequality

$$2R \geq t(t^4 + 7t^2 + 9)(A_1 + A_2)$$

and $\int_0^\infty 4/(t^4 + 7t^2 + 9) dt = 2\pi/(3\sqrt{13})$ give (11). The complete derivation and its 1290-term exact coefficient check are [14, 15]. $\square$

Join cyclic blocks when they share a cut vertex; the components are *shared-cut clusters*. In the block-cut tree contract each cluster, take the minimal subtree spanning the resulting marked cluster nodes, and suppress only unmarked degree-two nodes. Components off this cyclic hull are trees with a unique hull attachment and stay with that attachment’s owner. The resulting *reduced cluster tree* has only marked leaves. A reduced edge records a nonempty chain of actual bridges; it is never treated as an analytic edge.

Lemma 2.3 (Cycle order and interval realization). *Let a cycle C receive one, two, or three owner marks from components of the graph outside C . Their cyclic order determines a partition of C into nonempty proper consecutive vertex intervals, with coincident marks retained together. Assigning each open arc and either boundary vertex according to that order gives disjoint induced territories. A later router operation may refine one existing interval but never crosses an earlier boundary.*

Proof. Delete the edges of C and inspect the components rooted at its vertices. Each has one root, so its owner is forced. For two distinct marks, cut one edge in each complementary arc; for three marks, cut one edge in each of the three arcs. Move each cut within its open arc until every interval is nonempty. The cactus property excludes any route between two such components outside C , so the resulting vertex sets are induced. Refinement repeats the same construction inside one interval and therefore preserves all earlier boundaries. $\square$

Lemma 2.4 (Global ownership). *Suppose connected subtrees of the reduced cluster tree are assigned to packet owners and every selected router is split as in Lemma 2.3. Then the assignments lift to an induced, disjoint, exhaustive vertex partition of the original cactus. Every bridge remnant, connector entry, shared cut, incidence branch, and off-hull tree has exactly one final owner. If several router operations are sequential, replacing each provisional owner when its interval is refined gives the same unique final ownership.*

Proof. Cut only actual bridges between reduced-tree territories and use the unique path property to assign every connector vertex. Inside a cluster apply Lemma 2.3; a branch at a cut follows that cut, while a tree at a private cycle vertex follows its interval. These rules cover every vertex and agree at every boundary. Induction on refinements proves the final-owner claim. $\square$

3 The exact rank-ten frontier

For an edge-containing graph put

$$\kappa(G) = \sup_{M \geq 0, \mathbf{1}^T M \mathbf{1} > 0} \frac{4(\sum_{uv \in E(G)} \sqrt{M_{uv}})^2}{\mathbf{1}^T M \mathbf{1}}.$$

The sharp cactus DNN theorem [3, 2] gives

$$\kappa(G) = b + \sum_i \kappa(C_{\ell_i}), \quad \kappa(C_\ell) = \begin{cases} \ell, & \ell \text{ even}, \\ \ell \sec^2(\pi/(2\ell)), & \ell \text{ odd}, \end{cases} \quad s^-(G) \leq \kappa(G).$$

Set $\epsilon_\ell = 0$ for even ℓ and $\epsilon_\ell = \ell \tan^2(\pi/(2\ell))$ for odd ℓ . Block counting gives

$$\sigma(G) \geq 9 - \sum_{i=1}^{10} \epsilon_{\ell_i}. \quad (12)$$

The odd sequence decreases, $\epsilon_3 = 1$, and with $a = \epsilon_5 = 5 - 2\sqrt{5}$ one has $3a < 2$, $2a > 1$, and $\epsilon_5 + \epsilon_7 < 1$. If k cycles are nontriangles, a nonpositive margin forces $k(1 - a) \leq 1$, hence $k \leq 2$. For $k = 2$, the last comparison leaves only two pentagons. Therefore the exact frontier is

$$T^9Q \quad (q \geq 3, \text{ including } Q = T), \quad T^8PP. \quad (13)$$

Both genuinely remain: their DNN margins are $-\epsilon_q$ and $1 - 2\epsilon_5 = 4\sqrt{5} - 9$.

The exact colored cluster census [5] is

residual	all partitions	proper	direct	structural
T^9Q	97	96	92	4
T^8PP	181	180	170	10

Here a cluster color is its triangle count together with its number of distinguished cycles. Every direct row has a positive exact packet ledger. The structural rows are

$$\begin{aligned} T^9Q: & \quad Q|T|T|T|T|T|T|T|T, \quad Q|T|T|T^7, \quad Q|T|T^8, \quad Q|T^9; \\ T^8PP: & \quad P|P|T|T|T|T|T|T|T|T, \quad P|P|T|T^7, \quad P|P|T^8, \\ & \quad P|T|T|T|T|T|T|T^2P, \quad P|T|T|T|T|T|T^3P, \\ & \quad P|T|T|T|T|T^4P, \quad P|T|T|T|T^5P, \quad P|T|T|T^6P, \\ & \quad P|T|T^7P, \quad P|T^8P. \end{aligned}$$

Lemma 3.1 (Exact pruning of the ten T^8PP structural rows). *For every topology of the reduced cluster tree, the ten structural rows in the T^8PP census have the following exhaustive ordered reductions. An all-triangle leaf is used first; its triangle count is at most eight and its complement has rank at least two.*

	colored row	outcome if no all-triangle cluster is a leaf
1	$P_0 P_1 8T$	$TP+$ connected rank eight
2	$P_0 P_1 T A_7$	$TP+$ connected rank eight
3	$P_0 P_1 A_8$	$P_0 A_8 P_1$
4	$P_1 6T T^2P_0$	$TP+$ connected rank eight
5	$P_1 5T T^3P_0$	$TP+$ connected rank eight
6	$P_1 4T T^4P_0$	$TP+$ connected rank eight
7	$P_1 3T T^5P_0$	$TP+$ connected rank eight
8	$P_1 2T T^6P_0$	$TP+$ connected rank eight
9	$P_1 T T^7P_0$	$TP+$ connected rank eight
10	$P_1 T^8P_0$	$T^8P_0 P_1$

Every displayed separation is made on actual bridges and lifts to disjoint, induced, exhaustive territories with a unique owner for every connector remnant and attached tree.

Proof. If an all-triangle cluster A_r is a leaf, cut the first actual bridge on its unique reduced-tree chain. Equation (7) makes that territory strict, and its connected complement has rank $10 - r \in \{2, \dots, 9\}$, hence is nonnegative (and is strict from rank four onward).

Assume there is no such leaf. Every leaf then contains a pentagon. Since there are only two pentagons, the reduced tree has exactly two leaves and is a path, with one pentagon at each end. In rows 1, 2, and 4–9, one end is the singleton P_1 and its adjacent marked cluster is a singleton triangle. Their terminal segment is a connected TP territory; cutting the first actual bridge on its other side leaves a connected rank-eight cactus. Both territories are strict. In row 3 the only remaining path is $P_0|A_8|P_1$, with A_8 between the two pentagons. In row 10 the unique path is the marked last-bridge endpoint $T^8P_0|P_1$.

For completeness, reduced trees with k marked clusters have only unmarked Steiner vertices of degree at least three and therefore at most $k - 2$ such vertices. Adding a marked leaf either at a vertex or by subdividing an edge generates every reduced tree inductively. The exact colored topology program enumerates all 156263 resulting classes in the ten rows and verifies that 156253 use the leaf alternative, eight use the terminal TP alternative, and one uses each marked endpoint [11]. Thus no path, Steiner, or high-degree disconnected topology is omitted.

On every bridge cut, each chain remnant follows its side, and every off-hull tree follows its unique attachment. Thus Lemma 2.4 applies. The ten rows are exhausted. A detailed graph-level audit is [10]. $\square$

Proposition 3.2 (Global cluster synthesis). *Every proper partition in the two residual families is strict. After direct packet rows and reduced-tree pruning, the only finite endpoints requiring a marked certificate are*

$$A_9 | Q, \quad T^8P_0 | P_1, \quad P_0 | A_8 | P_1. \quad (14)$$

Proof. For each of the first three T^9Q structural rows, every reduced tree has a triangular leaf, since all leaves are marked and Q is the only nontriangular cluster. Cutting its first actual bridge gives a strict triangular territory; the connected complement has rank at most nine and is nonnegative, with a strict packet on one side in every case. The fourth row is exactly $A_9 | Q$. Thus all Steiner topologies are exhausted. For T^8PP apply Lemma 3.1; its row table proves every structural topology or leaves exactly $T^8P | P$ and $P | A_8 | P$. Together with the 92 and 170 direct packet rows this exhausts the four and ten frozen structural rows. The marked closures of the three endpoints are proved next. $\square$

4 The marked disconnected endpoints

4.1 The endpoint $A_9 | Q$

Project the actual connector to its first shared cut or private triangle port. The exact marked-entry certificate [6] gives

355 unmarked A_9 incidences, 3624 = 3621+3 canonical marked classes, 6745 physical marked positions.

Every one of the 3621 direct classes has a legal triangle-router split with the TQ , TTQ , strict rank-three-through-nine, and triangular packet ledgers. If the entry is private, its interval remains with the real connector and Q .

Two residual marks are the common-cut nine-triangle bouquet, entered at the hub or privately; Lemma 2.2 or Lemma 2.1 gives $\sigma > 9 - \delta_q$. In the third, a triangle router joins a seven-petal hub to one leaf triangle and the entry is private on the router. Keep the connector with the router. The retained eight triangles share the real hub and the rooted Q , so Lemma 2.1 gives $\sigma > 8 - \delta_q > 0$. Thus all 3624 classes close. The executable freezes the three marked signatures and their orbit multiplicities 1, 1, 18, checks $6725 + 20 = 6745$ before quotienting, and assigns every connector, cut, interval, and attachment a final owner.

4.2 The last bridge $T^8P \mid P$

The exact certificate [8] reports

$$2392 \text{ } T^8P \text{ incidences, } 1105 \text{ with } P_0 \text{ an incidence leaf, } 11689 = 11586 + 100 + 3.$$

The 100 finite replacements use zero, one, two, and three routers in 2, 9, 73, 16 classes. Final-owner validation is part of the executable. The three residual marks lie on one two-cut kernel: a router triangle joins a common-cut A_7 fan to P_0 . Depending on the entry, the repairs are

$$A_7 + PP > 0, \quad P + \text{packing-one } A_7P > 7 - 2\delta, \quad \text{open } P_1 + \text{packing-one } T^8P_0 > 7 - \delta.$$

For the opening, the four private vertices of P_1 and all trees rooted there form a nonempty induced tree of surplus -1 ; the complement has the displayed eight-triangle common hub. All 11689 classes close.

4.3 The two-interface endpoint $P \mid A_8 \mid P$

The two labelled connectors may enter at cuts or private vertices and may coincide. Exhaustive placement on 126 unmarked A_8 incidences gives 36414 ordered placements and

$$11689 = 11674 + 15$$

canonical classes [9]. The best plans use zero, one, two, and three routers in 6, 10844, 838, 1 classes. The ordinary ledger is $c - e - 2\delta$, where c is the retained integer credit and e counts naked private-interface intervals; it accepts exactly the rows with $c - e \geq 1$.

Nine residuals are labelled forms of one two-hub kernel: a router R joins six petals at x and one leaf triangle at y . Its five surgeries give $A_6 + T + PP$, $A_6 + TP + P$, common-cut $T^6P + T + P$, or packing-one $T^6P + T + P$, with margins > 1 , $> 2 - 2\delta$, or $> 6 - 2\delta$. The other six are bouquet interface orbits. They close by opening one remote pentagon and retaining packing-one A_8P ; splitting an entered triangle to give $P+$ common-cut T^7P ; or producing $A_7 + PP$ or $A_6 + P + P$. The weakest margin is $1 - 2\delta > 0$. Lemmas 2.3 and 2.4 justify every labelled interval and connector owner.

This proves Proposition 3.2, hence every residual graph with more than one shared-cut cluster.

5 One fully shared cluster

5.1 The complete T^9Q census

The color-preserving incidence census [7] is

Q capacity	1	2	3	4	5	6	7	8	9	total
3	1	12	91	406	1178	2115	2250	1246	275	7574
4	1	12	91	412	1203	2187	2361	1340	306	7913
5	1	12	91	412	1208	2201	2393	1372	321	8011
6	1	12	91	412	1208	2204	2400	1383	327	8038
7	1	12	91	412	1208	2204	2402	1386	330	8046
8	1	12	91	412	1208	2204	2402	1387	331	8048
≥ 9	1	12	91	412	1208	2204	2402	1387	332	8049

Every nonhostile incidence except the common bouquet has an ordinary legal split. Actual hostility means $q \equiv 1 \pmod{4}$: among the displayed finite rows this includes $q = 5$, while $q = 7$ is nonhostile. The certificate also runs the weaker hostile ledger at capacity seven as a conservative check. For an actual hostile $q \geq 9$, the incidence universe is the saturated capacity-nine universe. In each of these weaker-ledger audits the exact frontier has the same three frozen signatures:

Q1 $X(Q()T()T()T()T()T()T()T())$

Q2 $T(X(Q()))X(T()T()T()T()T()T()T()T())$

Q3 $T(X(Q()))X(T())X(T()T()T()T()T()T()T()T())$

Q1 is common-cut and satisfies $\sigma > 9 - \delta_q$. In Q2 all nine triangles contain one hub and Lemma 2.1 gives the same bound.

Q3 is the essential firewall example. The saturated router meets Q , one leaf triangle, and a common-hub branch of seven triangles. The nine triangles do *not* have packing number one, so a global packing-one or Voronoi guard would be false. Open the leaf triangle: its two private vertices, their edge, and every tree rooted there form a nonempty induced tree E , so $\sigma(E) = -1$. The complement H contains the router and seven other triangles, all sharing the real hub, and Q rooted at the router's other cut. Therefore

$$\sigma(G) \geq \sigma(H) + \sigma(E) > (8 - \delta_q) - 1 = 7 - \delta_q > 0.$$

The executable checks all three signatures in each actual or conservatively hostile regime, canonical orbit sums, router degrees, the openable leaf, and the retained common hub. It materializes the two induced territories in Q3 and assigns every router cut and cycle attachment a final owner. Exact integer comparisons with $0 < \delta_q < 1$ certify the three strict margins, under both normal Python and `python -O`. Thus every fully shared T^9Q cactus is strict.

5.2 The complete T^8PP census: $30386 = 30377 + 9$

The exact census and ordinary split result are

cuts	1	2	3	4	5	6	7	8	9	total
all	1	19	204	1155	3990	8135	9615	5843	1424	30386
safe	0	17	200	1154	3989	8135	9615	5843	1424	30377
exceptions	1	2	4	1	1	0	0	0	0	9

The nine frozen signatures and replacements are:

code	c	replacement and positive margin
E1	1	common-cut T^8PP : $> 9 - 4/(3\sqrt{13})$
E2	2	open leaf P ; common-cut T^8P : $> 7 - \delta$
E3	2	$P+$ common-cut T^7P : $> 7 - 2\delta$
E4	3	open leaf P and leaf T ; common-cut T^7P : $> 5 - \delta$
E5	3	open one P ; packing-one T^8P : $> 7 - \delta$
E6	3	$P + T+$ common-cut T^6P : $> 6 - 2\delta$
E7	3	$P + P + A_6$: $> 1 - 2\delta$
E8	4	$P + P + T + A_5$: $> 2 - 2\delta$
E9	5	$P + P + T + T + A_4$: $> 3 - 2\delta$

Their signatures, in certificate order, are

E1 X(P)P(T)T(T)T(T)T(T)T(T)T(T))

E2 P(X(P))X(T)T(T)T(T)T(T)T(T)T(T))

E3 T(X(P))X(P)T(T)T(T)T(T)T(T)T(T))

E4 P(X(P))X(T))X(T)T(T)T(T)T(T)T(T)T(T))

E5 T(X(P))X(P))X(T)T(T)T(T)T(T)T(T)T(T))

E6 T(X(P))X(P)T(T)T(T)T(T)T(T))X(T))

E7 X(T)T(T)T(T)T(T)T(T)T(X(P))T(X(P)))

E8 X(T)T(T)T(T)T(T)T(X(P))T(X(P))X(T))

E9 X(T)T(T)T(T)T(X(P))X(T))T(X(P))X(T))

E5 is the second firewall row. Its ordinary split gives only $P + P + A_7$, whose lower ledger is negative. Open one leaf pentagon at cost -1 instead. The central router and seven hub triangles then form eight triangles with packing number one, joined to the other pentagon at the router's other vertex; Lemma 2.1 gives $> 8 - \delta$, hence $> 7 - \delta$ after opening. In E4, open the leaf pentagon and the leaf triangle at their private pivots, each at cost -1 , leaving common-cut T^7P and margin $> 5 - \delta$. The other rows use the displayed actual common cuts and legal triangle-router intervals. The nine-row verifier [12] regenerates all 30386 incidences, checks all signatures and edges, and certifies all nine exact ledgers.

6 Main theorem

Theorem 6.1. *Every connected cactus G of cyclomatic rank ten, on n vertices, satisfies*

$$\boxed{s^+(G) > n}.$$

Proof. The sharp DNN inequality (12) proves every cycle multiset except (13). For a residual, the reduced cluster tree either has more than one shared-cut cluster or one. In the first case, the exact $96 = 92 + 4$ and $180 = 170 + 10$ colored audits, Proposition 3.2, and the three marked endpoint certificates prove strict positivity. In the second, the complete T^9Q census and its three hostile closures, or the $30386 = 30377 + 9$ T^8PP census and nine replacements, prove it.

Every global cut is an actual bridge cut. Every cycle split follows cyclic order and uses nonempty proper consecutive intervals. Sequential operations refine one owner, and Lemma 2.4 gives every

connector remnant, cut, branch, and attached tree one final owner. Common-cut estimates are used only at displayed common pivots. Packing one is used only after direct verification of the retained hub; the two false-guard rows are instead opened at exact cost -1 . No open two-pivot statement enters. Hence $\sigma(G) > 0$, which is the assertion. $\square$

A Exact reproduction

The seven rank-ten programs require Python 3.10 or newer and use the standard library with exact integer or Fraction arithmetic. From the repository root run

```
python3 research/rank-ten-cactus-frontier-census.py
python3 -O research/rank-ten-cactus-frontier-census.py
python3 research/decacyclic-t9q-marked-entry-certificate.py
python3 -O research/decacyclic-t9q-marked-entry-certificate.py
python3 research/decacyclic-t9q-incidence-certificate.py
python3 -O research/decacyclic-t9q-incidence-certificate.py
python3 research/decacyclic-t8p-last-bridge-census.py
python3 -O research/decacyclic-t8p-last-bridge-census.py
python3 research/decacyclic-t8-two-interface-census.py
python3 -O research/decacyclic-t8-two-interface-census.py
python3 research/decacyclic-fully-shared-nine-exceptions.py
python3 -O research/decacyclic-fully-shared-nine-exceptions.py
python3 research/decacyclic-t8pp-reduced-tree-topology.py
python3 -O research/decacyclic-t8pp-reduced-tree-topology.py
```

Each program is run normally and under optimization. Its authoritative acceptance checks are explicit and fail closed; the dedicated marked and replacement programs check frozen totals, canonical classes or digests, interval ownership, router degrees, exact ledgers, and closure. Build this manuscript with

```
bash scripts/build-paper.sh all-decacyclic-cacti
```

References

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- [13] *Rooted packing-one hostile-cycle lemma for octacyclic provenance*, 26 July 2026; `research/octacyclic-packing-one-hostile-cycle-lemma-2026-07-26.md`.
- [14] *Common-cut residual bouquets: an exact rooted Schur–Sachs inequality*, 26 July 2026; `research/common-cut-bouquet-rooted-schur-2026-07-26.md`.
- [15] Exact coefficient verifier, `positive-square-energy/experiments/c5_bouquet_matching_certificate.py`.
- [16] *Two-pivot Schur–Sachs triangular cactus reduction*, open winding boundary, 26 July 2026; `research/two-pivot-schur-sachs-triangular-cactus-2026-07-26.md`.